Rocket Science and Technology 4363 Motor Ave., Culver City, CA 90232
Phone: (310) 839-8956 Fax: (310) 839-8855

Rocket Launches from a Stabilization Parachute(rev.1) 06 June 2013

by C. P. Hoult

Introduction

The problem of an air dropped sounding rocket is very interesting because both its initial conditions and its "rail" ending condition must be considered. There are two ways to launch a sounding rocket from an aircraft.

In the first, after being dropped from an aircraft, the rocket is allowed to glide nose first for a while. When well clear of the mother aircraft, a stabilization parachute is deployed from the rocket's nose. After the nose whips around the rocket is slowed while flying tail first. Gradually the parachute brings the rocket to a more nearly vertical attitude. When the rocket pitch attitude reaches its desired value, and the mother aircraft has adequate separation the first stage is ignited.

In the second approach, the mother aircraft pulls up into a very steep climb, and launches the rocket like any other missile. While there are clear performance benefits for the second approach, the cost and schedule required to certify the aircraft for launching this particular type of missile mitigate against it.

This memo focuses on the first approach. Like ancient Gaul, there are three parts to the analysis:

- First, the rocket's initial pitch attitude must be found.
- Second the effective launcher length must be obtained considering the launcher motion along its own length.
- Third, the distance traveled after ignition along a moving rail must be found.

Note that the launcher rail length in a well crafted point mass simulation never vanishes because of singular perturbation effects[1]. The rocket will appear to be constrained as though it were on a physical rail aligned with its initial attitude until the end of the "rail" is reached.

Initial Pitch Attitude

Consider a rocket dropped from a aircraft flying from right to left across the paper. Some time after stabilization parachute deployment, the system will appear as sketched below in Fig. 1. The sketch is accurate in that the rocket weight W pulls the rocket down away from streaming directly behind the stabilization parachute. The parachute drag D is in an opposite direction from the velocity V. If we include an deceleration term ma in the drag we can treat this as a statics problem. Neglect all aerodynamic forces except parachute drag because as the rocket becomes more vertical, these become increasingly less important due to slower speeds. Now, consider the balance of forces along the rocket's axis and normal to it:

$$(D - ma)\cos\psi - W\sin\theta = 0 \qquad (1)$$

$$(D - ma)\sin\psi + W\cos\theta = 0 \tag{2}$$

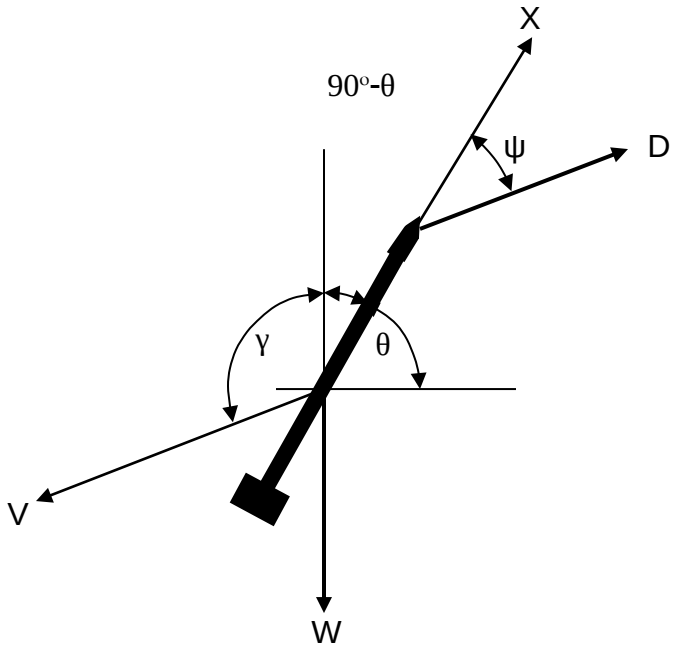


Figure 1 Rocket on Parachute Geometry

The forces can be eliminated from these two resulting in an angles only relation:

$$\sin\theta\tan\psi + \cos\theta = 0 \tag{3}$$

Next, geometry gives us that $\psi = 90^\circ - \gamma + \theta$ (4)

From here on, it's a small slog through the algebra. Substitute eq. (4) into eq. (3) to find that

$$\tan\gamma\tan^2\theta + 2\tan\theta - \tan\gamma = 0 \tag{5}$$

This is a quadratic equation with only one reasonable root,

$$\tan\theta = \frac{1 - \cos\gamma}{\sin\gamma} \tag{6}$$

During a simulation at any time step on the parachute γ is known, and, by virtue of eq. (6), the rocket's pitch orientation were it ignited at that time. Equation (6) is plotted in Fig. 2 below. If the rocket and parachute were initially traveling in a horizontal direction the rocket itself would droop 45°. And, if the parachute were descending vertically, the rocket would point straight up.

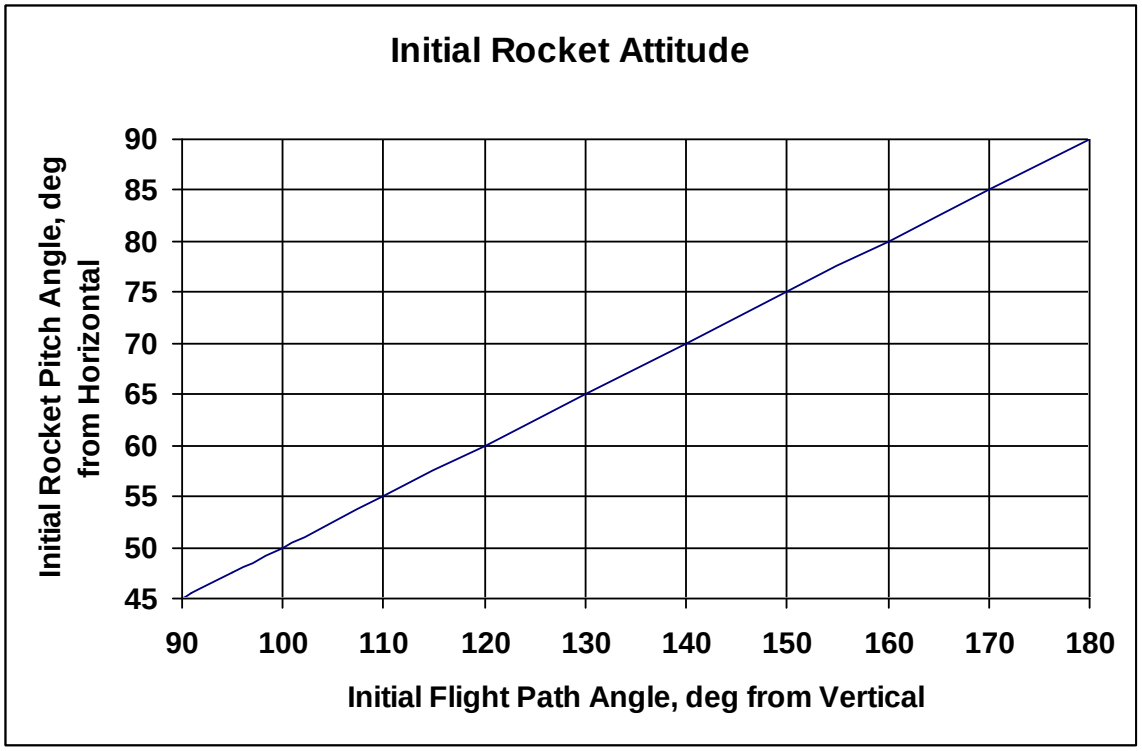


Figure 2. Initial Rocket Pitch Attitude

As a side benefit, if one times ignition carefully, it could be possible for the rocket to

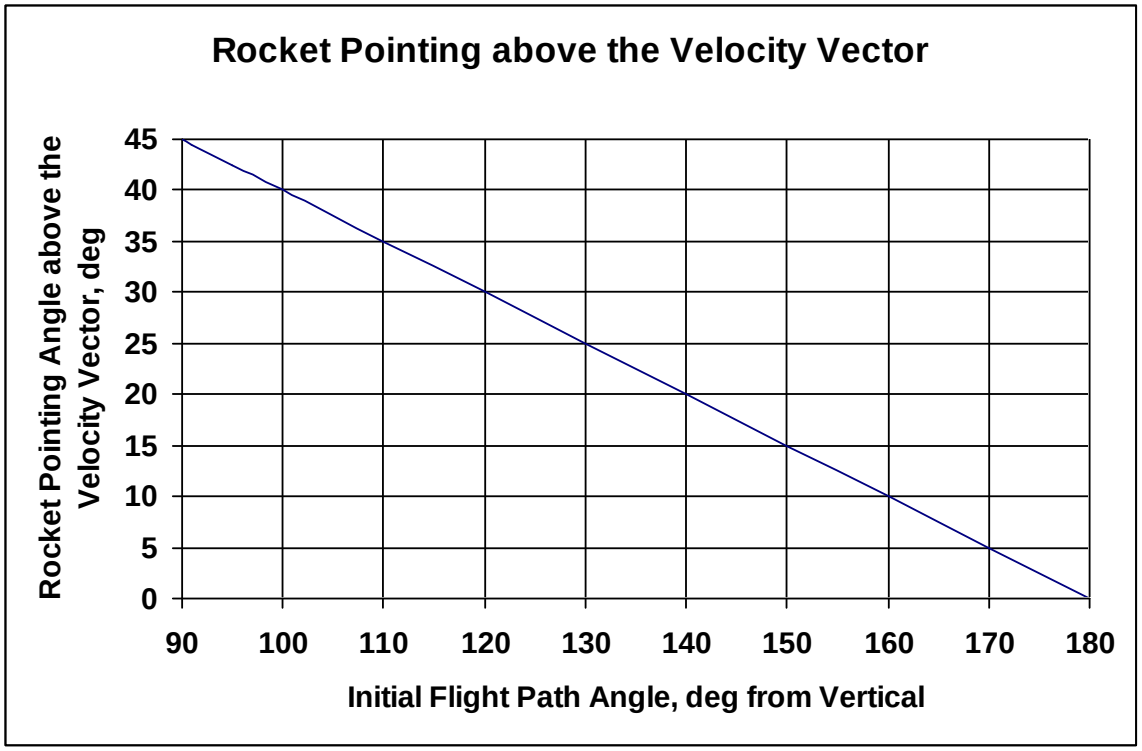


Figure 3. Initial Rocket Pointing Angle Above the Velocity Vector

miss the parachute canopy entirely. Figure 3. above shows the offset of the rocket's longitudinal axis relative to the initial velocity vector.

Finally, if the flight path angle γ, in radians, is redefined to be positive for rotations above the local horizon as is the practice in SKYAERO, the initial rocket Quadrant Elevation angle QE, in radians, is given by

$$QE = \pi/4 - \gamma/2 \tag{7}$$

This should be used in trajectory simulations.

Motion Along the Launcher Rail

The vast majority of air dropped/launched rockets do not actually fly from a physical launcher rail. How can the finite rail length developed in ref. (1) be reconciled with how a simulation is conducted? Let's consider a gedanken experiment. Suppose a rail of finite length is attached to the stabilization parachute. For inertia reasons, assume the rail's state of motion does not change during the time the rocket flies out along it. Because the rail is initially moving backward, it appears to shrink with time. That is, the rocket sees a launcher of apparent length L_{AP} at any time after ignition. Then if the rail length were L. and the speed on the parachute were V_0 then

$$L_{AP} = L - V_0 t \cos\psi \tag{8}$$

at any time t after ignition. Here the launcher length is the physical length plus the single perturbation increment from ref. (1). As the physical launcher length is made shorter and shorter eq. (8) continues to hold. Passing to the limit

$$L_{AP} = L_1 - V_0 t \sin(\gamma - \theta), \tag{9}$$

where L_1 is the ref. (1) value of the minimum simulation launcher length for the rocket in question.

The trajectory flown along a moving rail is, when viewed from a stationary frame, curved. But, the L_1s, for real world cases, are very short, on the order of one or two rocket lengths. Then, the flight path curvature of the trajectory on the rail can be neglected, and the distance Δs the rocket has flown after t seconds is approximately

$$\Delta s \cong \sqrt{(Alt - Alt_0)^2 + (Rng - Rng_0)^2} \tag{10}$$

Physically, what's happening as seen by an observer who does not move relative to the atmosphere is shown in Fig. (4) below. The dashed line is the path followed by the rocket.

To see what's quantitatively happening, consider a little constant acceleration model. The classical Galilean equations for motion along, and normal to, the rail when viewed by an observer fixed with respect to the atmosphere are:

$$x = at^2/2 - V_0 t\sin(\theta - \gamma), \text{ and}$$
$$z = V_0 t\cos(\theta - \gamma) \tag{11}$$

Here,
x = Displacement along the rail,
z = Displacement normal to the rail,
a = Thrusting acceleration along the rail,
t = Time after ignition,
V_0 = Velocity descending on the parachute,
θ = Rocket initial pitch attitude, and
γ = Initial flight path angle.

Take typical numbers to be;

$\gamma = 160^o$,
$a = 150$ ft/sec^2
$V_0 = 25$ ft/sec.

The results are shown in Fig's. (5) and (6) below:

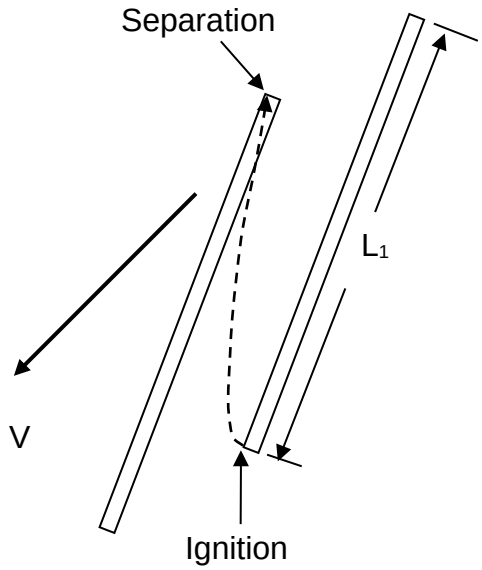


Figure 4 Rocket Trajectory on the Rail

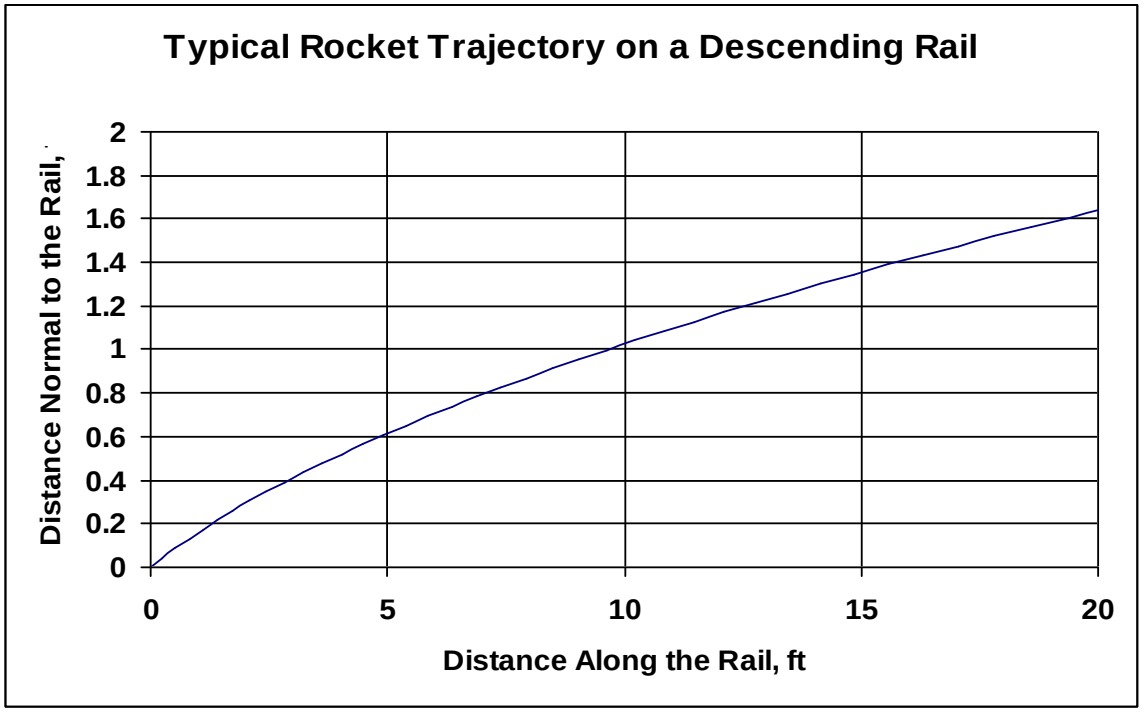


Figure 5 Rocket Trajectory on a Descending Rail

Next, compare the distance traveled as found by numerical integration along the curve of Fig. 5 with the approximation found from eq. (10):

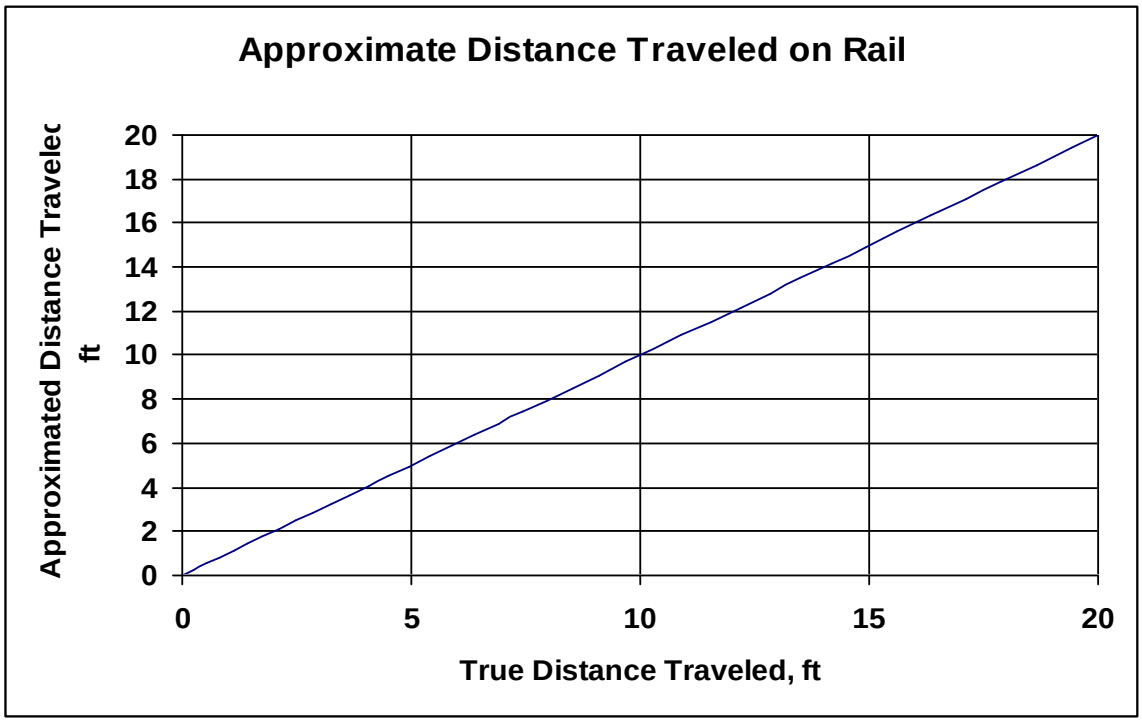


Figure 6 Distance Traveled Along the Rail

As can be seen, the approximation in eq.(9) should be plenty good enough.

When the initial velocity in the "QE" direction is negative (parachute is descending) select the simulation launcher length to be the sum of that needed to bring the velocity along the "rail" to zero plus an additional length L_1.

At any time after ignition, the velocity along the rail V is

$$V = \sqrt{V_{z0}^2 + V_{x0}^2}\cos(QE - \gamma_0) + \Delta V \qquad (12)$$

Here, the initial velocity components and flight path angle are in the notation used in SKYAERO. The change in velocity due to propulsion (and weight) is ΔV.

Summary

Let's recapitulate what has been laid our above. While the rocket descends on its stabilization parachute it awaits a time signal to ignite. When that happens, the initial flight path angle is obtained from eq. (6). The simulation launcher length is obtained from ref. (1), and the rail constrained equations of motion are integrated in the usual way. The phase boundary between rail-constrained motion and free flight is when Δs (eq.(10)) equals L_{AP} (eq. (9)).

Reference

1. "Launcher Length for Sounding-Rocket Point-Mass Trajectory Simulations", by C. P. Hoult, Journal of Spacecraft and Rockets, Vol. 13, No. 12, pp760-761